

Re: Response to Market Supporting Policy Subindex Evaluation

Dear Professor Pandya,

Thank you for your thorough and thoughtful evaluation of the Market Supporting Policy subindex. Your critiques are valuable, and we appreciate the care you took in reviewing this element of our Index. We have attached the full version of our GCI whitepaper, which was previously unavailable, and should add useful context to the value added by the project. We have also made substantive changes and additions to this subindex based on your recommendations and hope they sufficiently address your concerns. Where we made no changes, we provide a detailed explanation. Below are our responses to your feedback and recommendations.

I. Introductory Comments

“I recognize the limits of existing capitalism measures and appreciate the logic of emphasizing that institutions support capitalism. That said, I am not yet convinced that this index adds sufficient value for its range of possible users. Regarding academic researchers, can you show that switching to this index changes anything tangible such as research findings? For example, you can replicate statistical analyses replacing this index with existing measures. I do not typically use these types of measures myself, but I assume researchers use them in statistical analyses of country-year outcomes. Since country scores probably do not vary much annually, these would have to be analyses with a relatively long time series to capture meaningful within country variation over time.”

We agree that the index must demonstrate an empirical improvement relative to existing measures (Heritage, Fraser) to be useful for academic researchers.¹ The first and perhaps most important value added offered by the GCI is absolute transparency. The GCI provides:

- Raw source data
- Step-by-step data transformations with documented processes and formulas
- Published peer reviews for each subindex and the aggregate GCI, along with detailed responses outlining our alterations to the GCI theory and computation
- Technical appendices detailing every step in our technical analysis process
- Downloadable R replication packages, a comprehensive listing of dataset reliability grading criteria and scores (including all 6 reliability component scores and their weights), and a 150-page whitepaper that provides a detailed articulation of our theory for each subindex and the GCI as a whole

Neither Fraser nor Heritage offer any of these, thus putting severe limitations on scholars’ ability to test and assess the structural consistency and reliability of their results.

¹We would expect some natural overlap with existing economic freedom indices due to some shared concepts and datasets. In our preliminary analysis, the GCI plot against The Heritage Foundation’s Index of Economic Freedom and the Fraser Institute’s Economic Freedom Index yield the following correlations: GCI to Fraser at 0.748, GCI to Heritage at 0.722, and the two economic freedom indices against each other higher at 0.87. This illustrates the Global Capitalism Index’s differentiation from these existing indices and, simultaneously, an expected degree of similarity. Further detail on GCI scores and comparative plots can be found in the whitepaper.

Second, the GCI generates confidence intervals through bootstrap resampling, providing measures of uncertainty around index estimates. Neither Fraser nor Heritage provide critical estimates of uncertainty for researchers to properly adjust their findings. Additionally, robust standardization and winsorization procedures reduce sensitivity to extreme values that can distort final estimates.

Beyond its statistical advantages, the GCI differs conceptually from existing measures by measuring capitalism itself rather than “economic freedom”. As a result, the GCI captures institutional architectures, not solely the level of market liberalization, and presents subindices structured around each capitalist institution as independently operable entities. This means that, for the first time, academic researchers have both an aggregate index of global capitalism for simple comparisons (akin to V-DEM’s aggregate indices of democracy), as well as eight subindices for specialized research. In the analysis section of the GCI whitepaper, we provide correlative validation of this differentiation and use documented historical events (financial crises, political regime shifts, etc.) to compare how each index captures and predicts the direction and magnitude of change.

Aggregate vs. Component Metrics

“Related, when would researchers prefer a broad summary metric of capitalism over directly measuring relevant components? This index summarizes several economic and policy concepts with well-established and readily available metrics. For all the emphasis on a principled measure, I do not see a case for using this measure over precisely measuring just its relevant elements separately. Note that democracy measures (V-Dem, Polity Scores) are often an index, and they summarize concepts that are otherwise difficult to measure.”

If a researcher treats capitalism as a system of institutions rather than a collection of separable policies, it becomes crucial to measure components with coherence. Measuring individual components in isolation, though perhaps easier to draw out causal inference, implicitly assumes independent effects and dismisses the political-economic reality of complementarities.

The GCI addresses this gap by providing both a total system-level measure (aggregate index) and analytically separable subindices of its institutions. The aggregate index enables cross-national and intertemporal comparison when capitalism itself is the object of explanation, while the subindices allow researchers to test specific mechanisms and causal pathways within a unified framework. This structure preserves analytical flexibility without sacrificing conceptual coherence.

As you correctly identify, democracy indices like V-Dem and Polity are important measures of concepts that cannot be measured directly and are meaningfully understood as institutional regimes. The GCI applies this same logic to capitalism. The value of the index lies precisely in enabling regime-level measurement that existing, component-only approaches cannot provide.

Nonacademic User Value

“Non-academics may have greater demand for a summary measure in descriptive cross-country comparisons. The question of value added remains. Existing measure may be somewhat arbitrary but may be easier to understand. Non academic users could find the substantive focus on institutions valuable. The current version of the index does not lend itself to demonstrating the distinctive role of institutions.”

We agree that the GCI may have benefit outside of a purely academic audience if framed correctly. We are currently building a dynamic web-based interface with the explicit intention of disseminating

the GCI for public consumption. Existing measures are easier to understand precisely because they obscure the reality of institutional mechanisms. We intend to demonstrate why countries differ, not just that they differ, and create a website that allows for clear analysis of institutions on growth, volatility, and policy outcomes. We would welcome your thoughts on how best to present the distinct role of institutions to non-academic audiences.

II. Overall Index

Component Duplication

“I am puzzled by the overall organization of the index, especially the duplication of concepts across subindexes. For example, Market Supporting Policy is its own subindex but is also a subset of New Business Formation and Growth (Government Supporting Policy). They use different datasets to measure identical concepts (example: firms’ perceptions of government R&D support and government R&D support). At best, the categories need sharper definition. I also wonder if the duplication might bias the overall index by inflating those dimensions of variation in the second order PCA.”

The GCI is intentionally structured around eight sovereign subindices, each designed to function as an independently usable measure of a distinct institutional dimension of capitalism. This design choice ensures that each subindex can support standalone empirical work, while the second-stage PCA process prevents this duplication from biasing the aggregate index score. Substantively, the Market-Supporting Policy subindex measures policies designed to build, grow, and sustain functioning market systems across sectors, whereas the Government Supporting Policy component within New Business Formation and Growth measures entrepreneurship-specific support from the public sector.

Data Biases and WEF Surveys

“I would like to see more reflection on the types of data used. For example, Market Supporting Policy is almost entirely based on WEF firm surveys. Firm surveys have the well-known limitation of selection bias because they sample surviving firms. In general, I suggest aiming for distinct types of data for each concept: observed government investments (government spending on education, R&D, etc), observed quality and quantity of government investment outcomes (educational attainment, roads per capita, patents), government policies (tariffs, regulations), private sector perceptions (firm surveys), private sector actions (borrowing, capital flows), aggregate private sector outcomes (loan defaults).”

“Topics could be added to the index. In principle, there is no constraint on the amount of data or number of measured used (computing resources are plentiful) though it could make it harder to organize concepts.”

The GCI is intentionally designed to prioritize institutional and policy inputs rather than economic or social outcomes, not because outcomes are unimportant, but because analytical clarity requires separating the structures of capitalism from the results they produce. Firm surveys, like the WEF Executive Opinion Survey, are used as perception-based proxies for institutional credibility and policy coherence. We deliberately exclude many of the outcome variables suggested because these are precisely the dependent variables the GCI is intended to help explain. The objective is to provide researchers with a theoretically grounded, multi-dimensional set of independent variables

capturing how capitalist systems are institutionally constructed, so they can then test how different configurations of capitalism shape outcomes such as inequality, innovation, financial stability, or cross-border capital flows. While the index architecture is modular and could accommodate additional measures, we treat the current eight subindices as an exhaustive framework for capturing the institutional foundations of capitalism itself.

Inequality and Redistribution

“Income equality and government (re)distribution. It pops up in the labor market index and there is some passing mention of taxation. This is substantive judgment call but worth considering if you think inequality harms or helps capitalism.”

We view income inequality and government redistribution primarily as outcomes rather than components of capitalism. Our interest is in testing how variations in capitalism relate to these outcomes rather than incorporating them directly into the GCI itself. We think it is likely that the type of capitalism is closely connected with certain types of inequality. We want to provide researchers with analytical tools to test the connection rather than presupposing a relationship by including it in our inputs.

Monetary Policy and Central Bank Independence

“Monetary policy. I see central bank independence and the Chinn-Ito Index, a measure of controls on cross-border capital flows. What about exchange rate volatility, currency reserves, and other dimensions of monetary policy related to macroeconomic stability?”

We treat monetary policy as context-specific responses rather than constitutive features of capitalist systems. Because policy choices are often endogenous to macroeconomic context and external constraints (BoP pressures, external shocks, precautionary protective measures), including them would risk conflating policy responses with underlying institutional efficacy. Central bank independence, capital account restrictiveness, and system liquidity are measurable institutional inputs across countries that define the rules, constraints, and capacities within which monetary authorities operate, rather than the policy choices or macroeconomic outcomes that emerge.

“Public finance (government debt/GDP , sovereign debt ratings).”

We include country risk metrics (country risk equity premiums, political risk indicators) in several subindices (Capital Market Sophistication, Strength of Property Rights) because they capture relevant constraints on economic activity, but we treat other public finance measures, like debt-to-GDP, as outcomes rather than components of capitalism. This keeps the index focused on institutional capacity while still allowing researchers to test how our subindices correlate with broader fiscal and macroeconomic outcomes.

Political Risk Measures

“Policy stability/uncertainty. This concept is central to political risk measures. The basic idea is that policies are more stable in the presence of more veto players and longer time horizons. Some of the democracy measures may captures these concepts. See also Political Constraint dataset by Wit Heinsz at Wharton.”

We incorporate several indicators of state stability in other subindices (namely Strength of Private Property and Private Ownership), including World Bank Governance Indicator, the Fragile States

Index, and V-DEM’s measures of constraints on executive power and checks on the judiciary. We deliberately exclude broad measures of political and state stability from the Market-Supporting Policy subindex because they capture background constraints on state action, not policy choices or institutional investments aimed at creating and supporting markets.

The Political Constraint Index by Wit Heinsz is a comprehensive measurement and meets our country and temporal coverage thresholds for inclusion in the GCI. We have incorporated it into the Strength of Property Rights and Private Ownership subindex, as it measures the structural constraints on government actors and the predictability of policy, rather than active market-supporting interventions. Higher Political Constraint Index scores indicate stronger institutional checks and more stable policy environments, which help safeguard property rights and reduce arbitrary government interference.

Other Useful Datasets

“Other possibly useful datasets: PRS country risk ratings, BERI ratings, World Bank Enterprise Surveys. The OECD has several index-like measures (example: Services Trade Restrictiveness Index) but coverage is obviously limited to the OECD and select other countries.”

Though PRS and BERI ratings would be beneficial additions, we intend for the GCI to be publicly available and replicable. The proprietary nature of PRS and BERI ratings prohibits us from using and publishing raw data for the purposes of replicability and reliability testing. We see significant merits to both approaches, and while we are airing on the side of transparency and replicability for this inaugural GCI model, we are not close to including proprietary data in the future.

We do include the OECD STRI in our Free Flow of Goods and Capital subindex, but, in general, seek to limit our reliance on OECD indices which, as you correctly point out, are limited by country coverage. Including OECD data tends to require more data imputations and widen our confidence intervals. We include the STRI in the Free Flow of Goods and Capital because it is a clearly defined, policy-based input measure that codes formal regulatory restrictions on services trade rather than outcomes. Given the scarcity of globally comparable, input-oriented measures of services trade policy, the STRI provides unique analytical value despite its limited coverage.

III. Market-Supporting Policy Comments

“Conceptually, I am not sure why this is a separate subindex. It is thin (relatively few measures compared to other subcategories) and overlaps a lot with other parts of the index. Perhaps consider whether all policy choices should be within this category. Alternately, consider whether this category is really about government provision of public/collective goods that support capitalism: infrastructure (paved roads per capita, (air)port capacity, average time to clear cargo), utilities (cost, reliability, availability of water, electricity, communications), health spending. Analogous education measures are already included elsewhere.”

We agree that the Market-Supporting Policy subindex is a unique element of the broader Global Capitalism Index. We also are aware that it contains fewer measures than some other subindices, (though it does rank 5 of 9 in terms of the number of dataset inputs). As such, we have considered many solutions, including combining this subindex into the Market Competition subindex to broaden both categories and widen the combined base of data.

Nevertheless, we view Market-Supporting Policy as a distinct and theoretically essential element of the broader GCI. We believe that markets are not naturally occurring, and that public and private actors must provide a wide architecture for markets to emerge and function. This subindex is designed to measure that architecture. Put another way, we see this subindex as precisely, and only, measuring the institutional and policy frameworks that supports the creation of markets themselves, as opposed to the activity that happens within those markets after they emerge. Once markets come into being and are sustainable, a host of actions can influence, affect, and alter the activities within those market systems. However, the market systems themselves must first be created and operate reasonably well. This subindex aims to measure the foundational institutional constructs of market systems, not market activity. For this reason, we do not believe that a merger between Market-Supporting Policy and any other subindex will adequately capture obscure but essential actions that create a conducive environment for the emergence and operation of market systems.

Of course, the theoretical necessity of Market-Supporting Policy remaining its own, independent metric does not address the relative “thinness” of its data base. As such, we are incredibly grateful for your recommendations. We will add/include multiple infrastructure metrics (such as road mileage per capita and [air]port capacity), as we believe that well-functioning linkages between centers of physical, financial, and human capital are critical to the very essence of market creation and operation. While slightly more problematic, we agree that the provision of certain utilities like electricity and communications are essential to well-functioning and operating market systems. However, we find health spending and education to be more peripheral. While they are obviously essential for human prospering, we are not certain that they in and of themselves are vital for the development and sustenance of market systems. As such, we argue that the development of human capital is less about the architecture of market systems than the operations and activities within those systems.

We will also include our proprietary “Hamilton indicator,” which proxies a government’s ability to use commercial and trade policy to create markets for essential goods while maintaining a competitive environment. Information on the Hamilton indicator can be found below.²

IV. Miscellaneous Comments

Pedagogical Applications

“Students learning basic principles of political economy could use something like this index to gain insight into big questions such as “why are some countries wealthy and others

²In establishing the fundamentals of the early American economy, Alexander Hamilton believed in striking a balance between economic self-sufficiency and an open trade policy to sustain capitalism in the long-run. His *Report on Manufactures* reflects this understanding of limited but energetic state interventions in trade policy to ensure domestic stability, and thus the ability of domestic firms to develop and produce goods and services for export, as well as helping “accelerate the transition to self-sustaining industrial capitalism.” (Douglas A. Irwin, “The Aftermath of Hamilton’s ‘Report on Manufactures’”) This logic is integrated into the subindex through the “Hamilton Indicator”, which yields high scores for low mean tariff rates with high levels of variation, weighted by a measurement of elite capture to account for the disequilibrium created by lobbying and policy objectives implemented without the public interest in mind. Mathematically, this indicator assigns positive values to lower mean tariffs rates in combination with higher coefficients of variation values. We use a 75% weight to mean tariff (WTO simple average MFN rate) and 25% weight to coefficient of variation (WTO coefficient of variation MFN). We then weight that value by V-DEM’s Particularistic vs Public Goods indicator (v2dlncmps) to proxy the potential for rent-seeking or rent-capturing behavior in policy formulation.

poor?” If that is a goal, consider how to tailor the index accordingly to be transparent and highlight the role of institutions.”

A core motivation of the GCI is to make the institutional foundations of capitalism legible to students and non-specialists grappling with classic political economy questions. For that reason, we emphasize transparent construction, clearly defined subindices, and publicly accessible underlying data. By decomposing capitalism into distinct institutional components, the GCI allows users to see which institutions matter, how they differ across countries, and how they change over time.

The index is deliberately constructed around these inputs so that users can also observe how different configurations of capitalist systems relate to downstream outputs like wealth, inequality, innovation, or political resilience. This pedagogical orientation will be reinforced through the GCI’s public web interface. In addition to transparent documentation and access to underlying data, the platform will allow users to explore correlations between individual subindices and major political economy outcomes. To your point, an interesting way to make these accessible to a larger audience could be creating a “big questions” section on the website where users can easily access the correlative relationship with a number of important issues.

Quantitative in the Age of AI

“Students and other lay audiences have a dangerous tendency to assume something numerical is precise and accurate. The advent of AI magnifies this tendency. Consider the implications of your index in this context and how it could be used to address these challenges.”

By publishing a transparent methodology, detailed confidence intervals, and all underlying data, we aim to produce the most trustworthy and replicable measure of global capitalism possible. Our goal is not to dictate how these numbers are interpreted, but to give students, researchers, and other audiences the tools to explore, test, and apply them responsibly. Additionally, a non-technical explanation of estimation uncertainty and methodology will be provided on the public-facing website.

Concluding Remarks

We are grateful for your detailed engagement with both the Market-Supporting Policy subindex and the broader Global Capitalism Index. Your comments pushed us to sharpen our theoretical justifications, address data limitations, and more clearly articulate the GCI’s value relative to existing measures. In response to your feedback, we have clarified the GCI’s focus on institutional and policy inputs rather than outcomes, reinforced the logic of the Market-Supporting Policy subindex as a theoretically distinct and necessary component, added infrastructure and utilities indicators to address concerns about the subindex’s relative thinness, and incorporated the Political Constraint Index into the Strength of Property Rights and Private Ownership subindex. We have also addressed concerns about WEF survey reliance, subindex overlap, and the index’s academic and non-academic utility. We would be happy to discuss any of these points further, and we hope to continue this productive dialogue about how to strengthen the Market-Supporting Policy subindex.

Sincerely,

UVA Democracy and Capitalism Lab